Name:

CSE4088 Introduction to Machine Learning

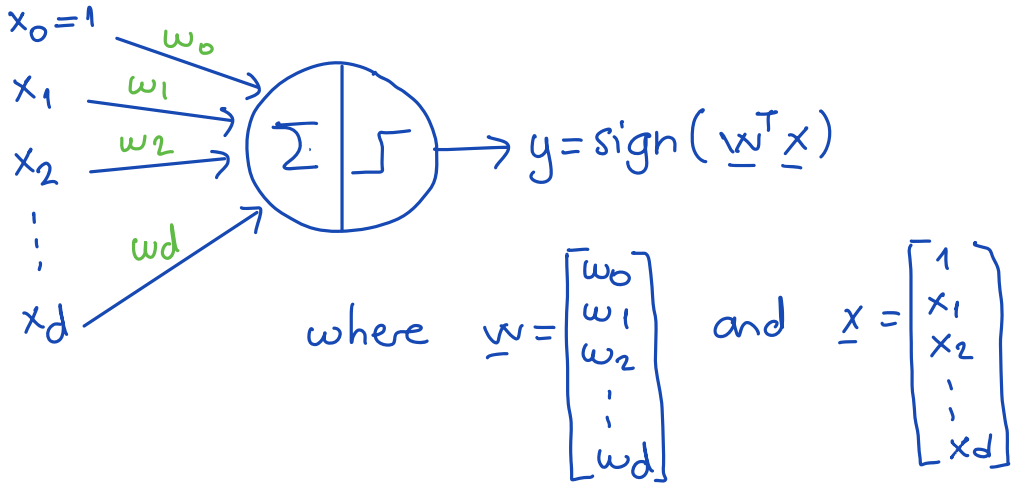
Quiz 1 – Solution

The Perceptron Learning Algorithm to classify positive (P) and negative inputs (N) is given below. Here x is a (d+1) dimensional vector with d features and the first element as 1. Similarly, w is a (d+1) dimensional vector

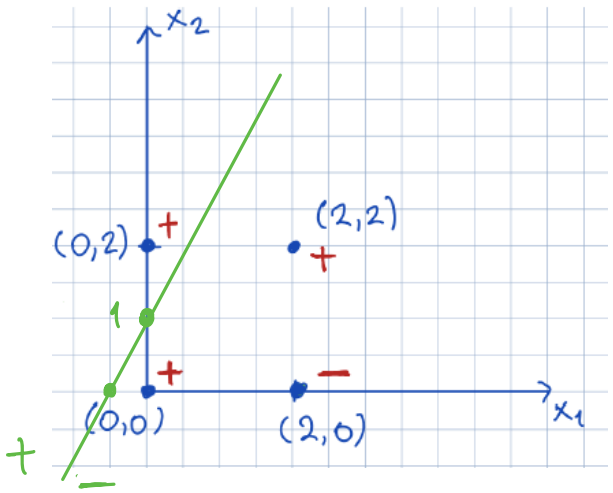
Algorithm: Perceptron Learning Algorithm

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P ← inputs with label 1;
N ← inputs with label 0;
Initialize w randomly;
while !convergence do
    Pick random x ∈ P ∪ N ;
    if x ∈ P and w.x < 0 then
        w = w + x ;
    end
    if x ∈ N and w.x ≥ 0 then
        w = w − x ;
    end
end
//the algorithm converges when all the
 inputs are classified correctly
```

a) Provide a schematic diagram of a simple perceptron neuron and describe mathematically its function.



b) Given d = 2, let the set P be the set of three points $(x_1, x_2) = \{(0,2), (0,0), (2,2)\}$ with positive labels and let N be the set of points $(x_1, x_2) = \{(2,0)\}$ with a negative label. The four points are shown below. Let the w vector be initialized as: $w = [w_0, w_1, w_2] = [-1 \quad -2 \quad 1]^T$. Draw the decision boundary (the line) on the figure below and indicate the positive and negative regions of the 2D space defined by the line.



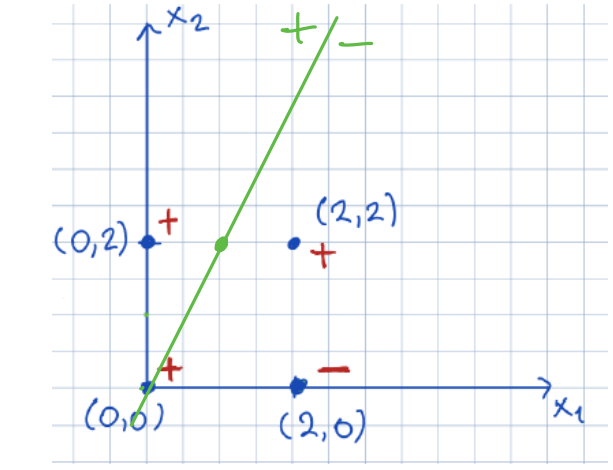
Line equation: $\underline{w}^T\underline{x} = 0$

$$[-1 \quad -2 \quad 1]\begin{bmatrix}1\\x_1\\x_2\end{bmatrix} = -1 - 2x_1 + x_2 = 0$$

Try $(x_1, x_2) = (0,0) \Rightarrow \underline{w}^T\underline{x} = -1 < 0$

negative region is below the green line

c) Indicate the misclassified points using the initial w. Perform one iteration of the PLA for one of the classified point(s) and update the w vector. Draw the new line on the figure given below. Indicate the misclassified points if any using the updated w.



Points (0,0) and (2,2) are misclassified since they are below the line:

w^Tx for $\underbrace{(0,0)}_{+}$ is: $[-1 \; -2 \; 1]\begin{bmatrix}1\\0\\0\end{bmatrix} = -1 < 0$

w^Tx for $\underbrace{(2,2)}_{+}$ is: $[-1 \; -2 \; 1]\begin{bmatrix}1\\2\\2\end{bmatrix} = -3 < 0$

Update w using the misclassified point (0,0):

$$\underline{w}_{new} = \underline{w}_{init} + \underline{x}$$
$$= [-1 \quad -2 \quad 1]^T + [1 \quad 0 \quad 0]^T = [0 \quad -2 \quad 1]^T$$

line: $-2x_1 + x_2 = 0$

Point (2,2) is still misclassified.

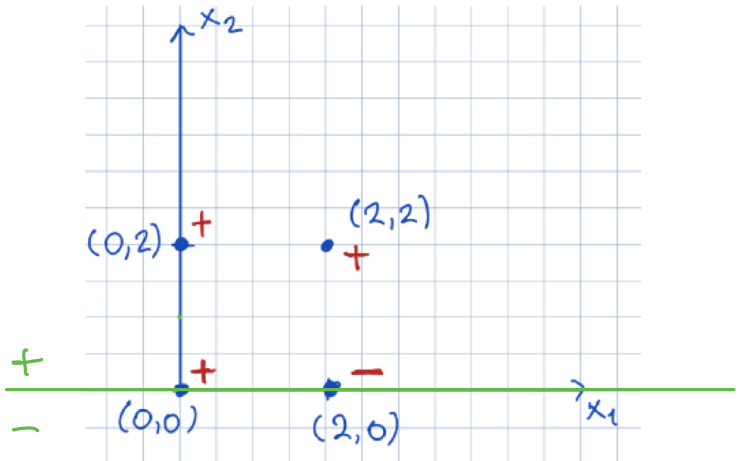
OR Update using the misclassified point (2,2)

$$\underline{w}_{new} = \underline{w}_{init} + \underline{x}$$

$$= [-1 \ -2 \ 1]^T + [1 \ 2 \ 2] = [0 \ 0 \ 3]$$

$$3x_2 = 0$$

$$x_2 = 0 \text{ line}$$



Point (2,0) is misclassified.